MILITARY SPECIFICATION
VALVE AND AIR CYLINDER ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criterial for one type of cylinder assembly entitled as Valve and Air Cylinder Assembly which is used for training on the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8821913	- Valve and Air Cylinder Assembly.
8822379	- Valve Assembly Training.
EL-8842322	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The valve and air cylinder assembly shall comply with all requirements specified on the drawing (dwg) 8821913, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Environmental testing. The valve and air cylinder assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The valve and air cylinder assembly shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.4.1 Vibration. The valve and air cylinder assembly shall show no evidence of mechanical damage such as visible loosening or deformation of any of the parts and shall be capable of retaining a pressure of 15 pounds per square inch (psi) when tested as specified in 4.4.3.

3.5 Workmanship.

3.5.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.3 Burrs. Parts shall be free from burrs.

3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contractor requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 8821913) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Vibration (see 3.4.1)		X
3. Overpressure (see 4.3.3.1)		X
4. Leakage (see 4.3.3.1)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Valve assembly training, prior to assembling needle valve, spring and plug (see dwg 8822379).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread	Gage
102.	Valve seat damaged	Visual
103.	"O" Ring damaged	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.5)	Visual

4.3.2.1.2 Valve and air cylinder assembly (see dwg 8821913).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Threads damaged	Visual
102.	Marking incorrect, mislocated or unidentifiable	Visual
103.	Any component loose or missing	Visual/Manual
104.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Protective finish damaged to the extent of exposing base metal	Visual
202.	Evidence of poor workmanship (see 3.5)	Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing in accordance with MIL-STD-1169	Visual

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102. Marking in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.5).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Overpressure	Major
b. Leakage	Major
c. Vibration	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Overpressure and leakage (see dwg 8821913). Each valve and air cylinder assembly shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with dwg 8821913, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. One sample valve and air cylinder assembly shall be selected at random from each lot of twenty-five (25) assemblies or portion thereof produced and subjected to the test (see 4.4.3). If any assembly fails to comply with 3.4, the lot shall be rejected. The first article sample will satisfy the first lot sampling requirement.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842322). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Overpressure. The assembly shall be connected to an air supply capable of producing a pressure of 22 psig. The pressure in the cylinder shall be slowly increased from 0 to 22 psig. The relief valve shall open when a pressure of 18 to 22 psi is reached, and closed and remain closed when 17 to 13 psi is reached.

4.4.2 Leakage. The assembly shall be filled with compressed dry air to 15 + 2 psig pressure. The pressure reading shall be recorded and the assembly shall be stored at ambient temperature for 24 hours. At the conclusion of the storage period a pressure reading shall be taken.

4.4.3 Vibration. The assembly shall be filled with air to 15 + 2 psig and secured to a vibration platform. A simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion) shall be applied, the frequency being varied uniformly between the approximate limits of 10 and 55 Hertz (Hz). The entire frequency range from 10 to 55 to 10 Hz shall be transversed in approximately one minute. The motion shall be continuously applied for 15 minutes in each of the three mutually perpendicular directions. After completion of this test a pressure reading shall be taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The valve and air cylinder assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The valve and air cylinder assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The valve and air cylinder assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be prepared in accordance with MIL-STD-1169(MU).

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1817 Amendment 1 dated 14 September 1966.

Custodian:
Army - PA

Preparing activity:
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